

LAPORAN PRAKTIKUM PERANCANGAN JARINGAN

Pemilihan dan Penerapan Protokol Switching dan Routing #4



Disusun oleh

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**PROGRAM STUDI DIPLOMA IV TEKNOLOGI REKAYASA INTERNET
DEPARTEMEN TEKNIK ELEKTRO DAN INFORMATIKA SEKOLAH VOKASI
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TUJUAN

- a. Memahami perancangan jaringan secara *top-down*
- b. Menerapkan rancangan jaringan berhirarki
- c. Melakukan konfigurasi protokol-protokol
 - VLAN
 - Subnetting dan Pengalamatan IP
 - Inter-VLAN Router
 - DHCP
 - SSH
 - OSPF
 - Switching Security
 - Host
- d. Pengujian dan Verifikasi komunikasi jaringan

DASAR TEORI

Faktor-faktor yang dilibatkan dalam sebuah pengambilan keputusan :

- Tujuan harus jelas
- Pilihan-pilihan atau ide yang ada harus dipelajari lebih jauh terlebih dahulu.
- Konsekuensi dari setiap keputusan harus dipelajari lebih lanjut
- Rencana cadangan harus segera dibuat.

Selain faktor-faktor diatas juga harus melibatkan beberapa pertanyaan berikut ini :

- Jika hasil keputusan yang diambil ternyata salah, apa yang akan dikerjakan ?
- Jika hasil keputusan sudah pernah dilakukan sebelumnya, apakah ada permasalahan dengan keputusan tersebut ?
- Bagaimana reaksi pelanggan dengan keputusan anda ?
- Apa rencana cadangan anda jika keputusan anda ditolak oleh pelanggan ?

Berikut ini adalah salah satu contoh dokumen dalam pembuatan keputusan

Critical Goals				Other Goals		
	Adaptability— must adapt to changes in a large internetwork within seconds	Must scale to a large size (hundreds of routers)	Must be an industry standard and compatible with existing equipment	Should not create a lot of traffic	Should run on inexpen- sive routers	Should be easy to configure and manage
BGP	X*	X	X	8	7	7
OSPF	X	X	X	8	8	8
IS-IS	X	X	X	8	6	6
IGRP	X	X				
EIGRP	X	X				
RIP			X			

*X = Meets critical criteria. 1 = Lowest. 10 = Highest.

Pemilihan Switching Protocols

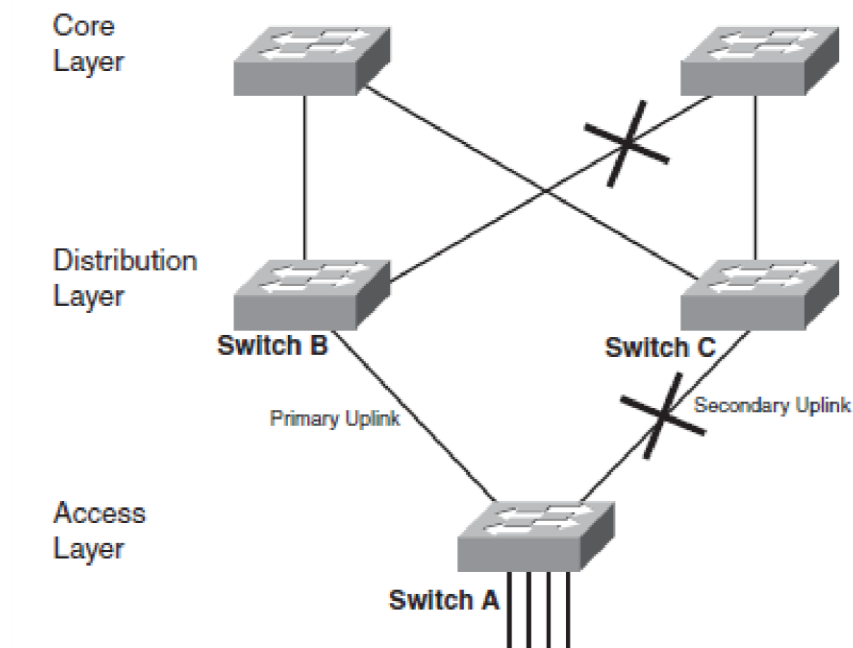
Switch memanfaatkan sirkuit terintegrasi cepat untuk menawarkan latensi rendah. Bridges jauh lebih lambat daripada switch dan memiliki lebih sedikit port dan biaya per port yang lebih tinggi. Untuk alasan ini, switch telah menggantikan bridges. Switch memiliki kemampuan untuk melakukan pemrosesan *store-and-forward* atau pemrosesan *cut-through*. Dengan pemrosesan *cut-through*, sebuah switch dengan cepat melihat alamat tujuan (bidang pertama dalam header frame Ethernet), menentukan port keluar, dan segera mulai mengirim bit ke port keluar. Kerugian dengan pemrosesan *cut-through* adalah meneruskan frame ilegal (misalnya, kerdil Ethernet) dan frame dengan kesalahan CRC.

Pada jaringan yang rawan kegagalan dan kesalahan, pemrosesan *cut-through* tidak boleh digunakan. Beberapa switch memiliki kemampuan untuk secara otomatis berpindah dari mode *cut-through* ke mode *store-and forward* ketika ambang kesalahan tercapai. Fitur ini disebut pengalihan *cut-through adaptif* oleh beberapa vendor. Switch memungkinkan beberapa jalur penerusan paralel, yang berarti sebuah switch dapat menangani volume lalu lintas yang tinggi lebih cepat daripada sebuah bridge. Switch high-end dapat mendukung banyak jalur penerusan simultan, tergantung pada struktur switching.

Berikut ini beberapa protokol switching yang dapat digunakan :

- STP
- RSTP
- PVST
- PVST+
- Etherchannel
- VTP
- DTP

Berikut ini contoh penggunaan STP



Gambar 1 Switch lapisan akses dengan dua uplinks ke switches layer distribusi
Pemilihan Routing Protocols

Protokol routing memungkinkan router secara dinamis mempelajari cara menjangkau jaringan lain dan bertukar informasi ini dengan router atau host lain. Memilih protokol routing untuk pelanggan desain jaringan Anda lebih sulit daripada memilih protokol switching, karena ada begitu banyak pilihan. Keputusan akan lebih mudah jika Anda dapat menggunakan tabel keputusan, seperti yang ditunjukkan pada Tabel 7-1 pada buku *Top-Down Network Design 3rd Edition*. Berbekal pemahaman yang kuat tentang tujuan dan informasi pelanggan Anda tentang karakteristik protokol routing yang berbeda, Anda dapat membuat keputusan yang tepat tentang protokol routing mana yang direkomendasikan. Berikut ini tabel ringkasan panduan dalam pemilihan protocol routing

Tabel 1 Perbandingan protokol-protokol routing

	Distance Vector or Link State	Interior or Exterior	Classful or Classless	Metrics Supported	Scalability	Convergence Time	Resource Consumption	Supports Security? Authenticates Routes?	Ease of Design, Configuration, and Troubleshooting
RIPv1	Distance vector	Interior	Classful	Hop count	15 hops	Can be long (if no load balancing)	Memory: low CPU: low Bandwidth: high	No	Easy
RIPv2	Distance vector	Interior	Classless	Hop count	15 hops	Can be long (if no load balancing)	Memory: low CPU: low Bandwidth: high	Yes	Easy
IGRP	Distance vector	Interior	Classful	Bandwidth, delay, reliability, load	255 hops (default is 100)	Quick (uses triggered updates and poison reverse)	Memory: low CPU: low Bandwidth: high	No	Easy
EIGRP	Advanced distance vector	Interior	Classless	Bandwidth, delay, reliability, load	1000s of routers	Very quick (uses DUAL algorithm)	Memory: moderate CPU: low Bandwidth: low	Yes	Easy
OSPF	Link state	Interior	Classless	Cost (100 million divided by bandwidth on Cisco routers)	A few hundred routers per area, a few hundred areas	Quick (uses LSAs and Hello packets)	Memory: high CPU: high Bandwidth: low	Yes	Moderate
BGP	Path vector	Exterior	Classless	Value of path attributes and other configurable factors	1000s of routers	Quick (uses update and keepalive packets, and withdraws routes)	Memory: high CPU: high Bandwidth: low	Yes	Moderate
IS-IS	Link state	Interior	Classless	Configured path value, plus delay, expense, and errors	Hundreds of routers per area, a few hundred areas	Quick (uses LSAs)	Memory: high CPU: high Bandwidth: low	Yes	Moderate

Berikut juga adalah tabel administrative distance berdasarkan tipe route untuk preferensi penentuan rute oleh router.

Tabel 2 Administrative Distance berdasarkan tipe route

Route Source	Default Distance Value
Connected interface	0
Static route	1
Enhanced Interior Gateway Routing Protocol (EIGRP) summary route	5
External Border Gateway Protocol (BGP)	20
Internal EIGRP	90
IGRP	100
OSPF	110
Intermediate System-to-Intermediate System (IS-IS)	115
Routing Information Protocol (RIP)	120
Exterior Gateway Protocol (EGP)	140
On-Demand Routing (ODR)	160
External EIGRP	170
Internal BGP	200
Unknown	255

Simpulan Pemilihan Protokol

Pemilihan protokol switching dan routing yang tepat untuk pelanggan desain jaringan Anda sangat diperlukan. Skalabilitas dan karakteristik kinerja protokol dan seberapa cepat protokol dapat beradaptasi dengan perubahan menjadi perhatian penting dalam pemilihan protocol witching maupun routing.

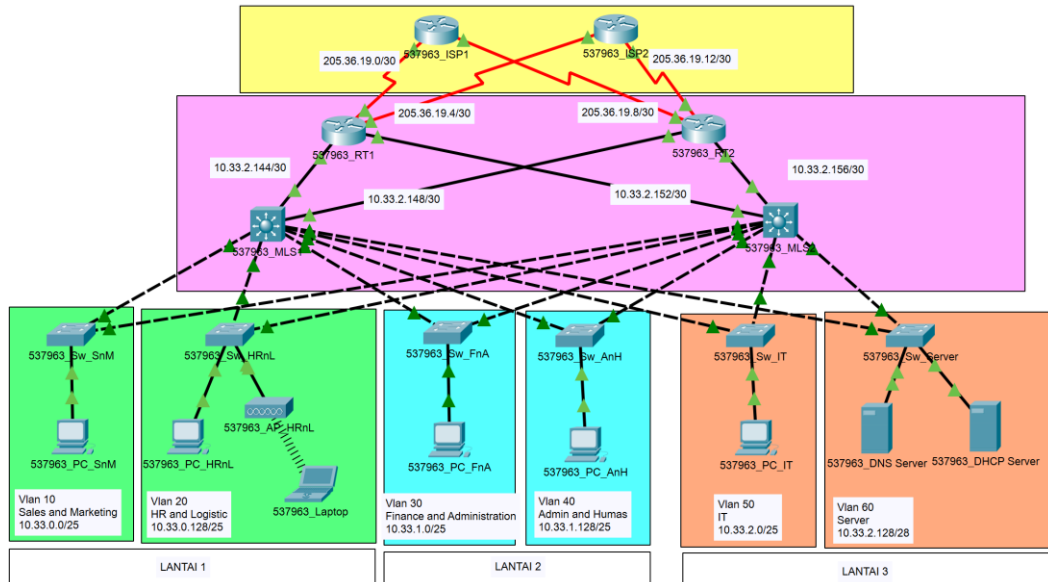
Memutuskan protokol switching dan routing yang tepat untuk pelanggan Anda akan membantu Anda memilih produk switch dan router terbaik untuk pelanggan. Misalnya, jika Anda telah memutuskan bahwa desain harus mendukung protokol routing yang dapat menyatu dalam hitungan detik di internetwork besar, Anda mungkin tidak akan merekomendasikan router yang hanya menjalankan RIP.

Detail pembahasan bagaimana memilih protocol switching dan routing sangaat direkomendasikan untuk membaca buku Top-down Network Design 3rd Edition dari CISCO yang juga sebagai buku utama matakuliah teori Perancangan Jaringan. Buku tersebut terdapat diskusi umum tentang pengambilan keputusan untuk membantu Anda mengembangkan proses sistematis untuk memilih solusi desain jaringan. Sebuah diskusi tentang menjembatani dan beralih protokol diikuti, yang meliputi menjembatani transparan, beralih multilayer, perangkat tambahan untuk protokol STP, dan VLAN. Bagian pada protokol routing mengikuti bagian switching.

PROSEDUR PERCOBAAN

HASIL PRAKTIKUM

1. Hasil topologi (lampirkan screenshot topologi dari packet tracer masing-masing).



2. Uji koneksi - Ping dari salah satu komputer ke komputer lain (**jelaskan dari perangkat mana ke perangkat mana**)

PC SnM ke PC IT

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.33.2.3

Pinging 10.33.2.3 with 32 bytes of data:

Request timed out.
Reply from 10.33.2.3: bytes=32 time<1ms TTL=127
Reply from 10.33.2.3: bytes=32 time=2ms TTL=127
Reply from 10.33.2.3: bytes=32 time=1ms TTL=127

Ping statistics for 10.33.2.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>ping 10.33.2.3

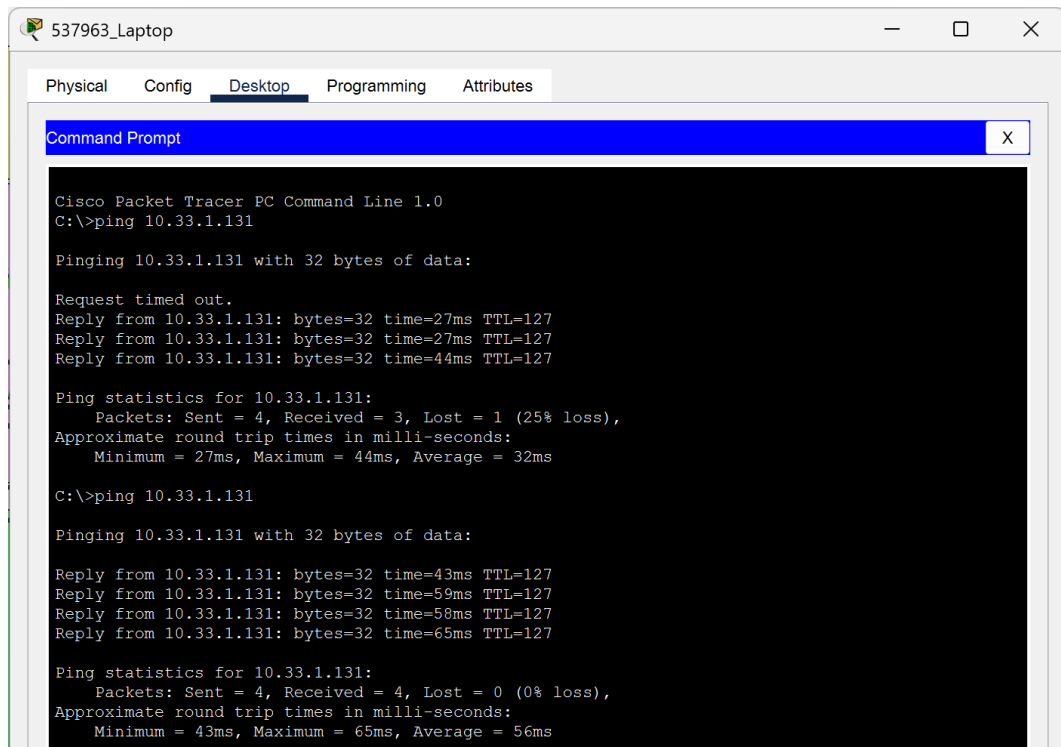
Pinging 10.33.2.3 with 32 bytes of data:

Reply from 10.33.2.3: bytes=32 time<1ms TTL=127
Reply from 10.33.2.3: bytes=32 time<1ms TTL=127
Reply from 10.33.2.3: bytes=32 time<1ms TTL=127
Reply from 10.33.2.3: bytes=32 time<1ms TTL=127

Ping statistics for 10.33.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

3. Uji koneksi wireless - Ping dari perangkat laptop ke salah satu perangkat lain.

Laptop ke PC AnH



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.33.1.131

Pinging 10.33.1.131 with 32 bytes of data:

Request timed out.
Reply from 10.33.1.131: bytes=32 time=27ms TTL=127
Reply from 10.33.1.131: bytes=32 time=27ms TTL=127
Reply from 10.33.1.131: bytes=32 time=44ms TTL=127

Ping statistics for 10.33.1.131:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 27ms, Maximum = 44ms, Average = 32ms

C:\>ping 10.33.1.131

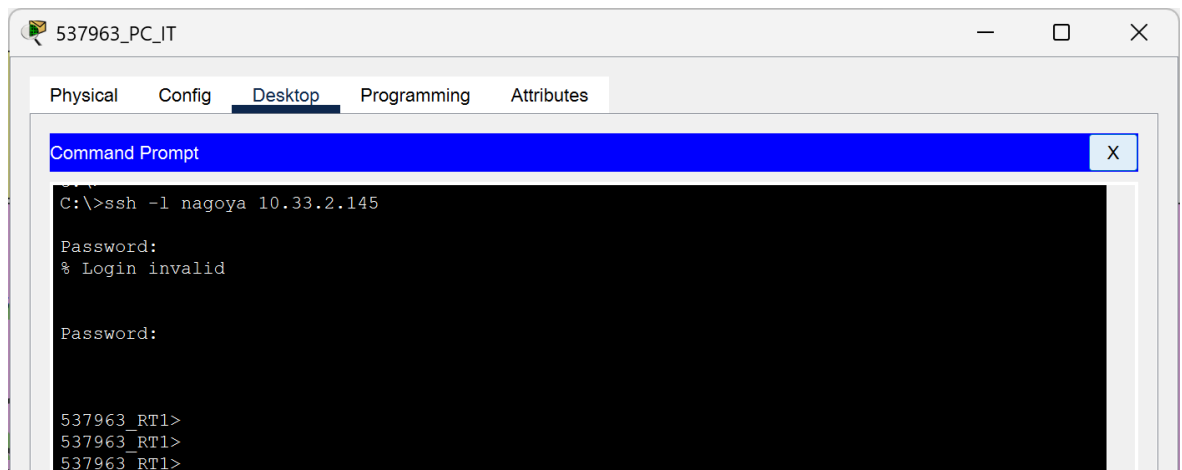
Pinging 10.33.1.131 with 32 bytes of data:

Reply from 10.33.1.131: bytes=32 time=43ms TTL=127
Reply from 10.33.1.131: bytes=32 time=59ms TTL=127
Reply from 10.33.1.131: bytes=32 time=58ms TTL=127
Reply from 10.33.1.131: bytes=32 time=65ms TTL=127

Ping statistics for 10.33.1.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 43ms, Maximum = 65ms, Average = 56ms
```

4. Uji SSH dari salah satu komputer ke salah satu perangkat router

PC IT ke router 1



```
C:\>ssh -l nagoya 10.33.2.145

Password:
% Login invalid

Password:

537963_RT1>
537963_RT1>
537963_RT1>
```

5. Verifikasi Interface

ISP 1

Device Name: 537963_ISP1					
Device Model: 2911					
Hostname: 537963_ISP1					
Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
GigabitEthernet0/0	Down	--	<not set>	<not set>	0060.7087.2601
GigabitEthernet0/1	Down	--	<not set>	<not set>	0060.7087.2602
GigabitEthernet0/2	Down	--	<not set>	<not set>	0060.7087.2603
Serial0/3/0	Up	--	205.36.19.1/30	<not set>	<not set>
Serial0/3/1	Up	--	205.36.19.9/30	<not set>	<not set>
Vlan1	Down	1	<not set>	<not set>	00E0.8FEE.3517
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 537963_ISP1					

ISP 2

Device Name: 537963_ISP2					
Device Model: 2911					
Hostname: 537963_ISP2					
Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
GigabitEthernet0/0	Down	--	<not set>	<not set>	0001.6357.6801
GigabitEthernet0/1	Down	--	<not set>	<not set>	0001.6357.6802
GigabitEthernet0/2	Down	--	<not set>	<not set>	0001.6357.6803
Serial0/3/0	Up	--	205.36.19.5/30	<not set>	<not set>
Serial0/3/1	Up	--	205.36.19.13/30	<not set>	<not set>
Vlan1	Down	1	<not set>	<not set>	0040.0BE8.87DB
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 537963_ISP2					

Router 1

Device Name: 537963_RT1					
Device Model: 2911					
Hostname: 537963_RT1					
Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
GigabitEthernet0/0	Up	--	10.33.2.145/30	<not set>	0005.5E38.6C01
GigabitEthernet0/1	Up	--	10.33.2.153/30	<not set>	0005.5E38.6C02
GigabitEthernet0/2	Down	--	<not set>	<not set>	0005.5E38.6C03
Serial0/3/0	Up	--	205.36.19.2/30	<not set>	<not set>
Serial0/3/1	Up	--	205.36.19.6/30	<not set>	<not set>
Vlan1	Down	1	<not set>	<not set>	0040.0B8E.9E77
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 537963_RT1					

Router 2

Device Name: 537963_RT2					
Device Model: 2911					
Hostname: 537963_RT2					
Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
GigabitEthernet0/0	Up	--	10.33.2.157/30	<not set>	0050.0FE4.9401
GigabitEthernet0/1	Up	--	10.33.2.150/30	<not set>	0050.0FE4.9402
GigabitEthernet0/2	Down	--	<not set>	<not set>	0050.0FE4.9403
Serial0/3/0	Up	--	205.36.19.10/30	<not set>	<not set>
Serial0/3/1	Up	--	205.36.19.14/30	<not set>	<not set>
Vlan1	Down	1	<not set>	<not set>	0005.5E08.5B71
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 537963_RT2					

MLS 1

Device Name: 537963_MLS1						
Device Model: 3650-24PS						
Hostname: 537963_MLS1						
Port	Link	VLAN	IP Address	IPv6 Address	MAC Address	
GigabitEthernet1/0/1	Up	1	10.33.2.146/30	<not set>	0001.64BC.DE01	
GigabitEthernet1/0/2	Up	1	10.33.2.149/30	<not set>	0001.64BC.DE02	
GigabitEthernet1/0/3	Up	--	<not set>	<not set>	0001.64BC.DE03	
GigabitEthernet1/0/4	Up	--	<not set>	<not set>	0001.64BC.DE04	
GigabitEthernet1/0/5	Up	--	<not set>	<not set>	0001.64BC.DE05	
GigabitEthernet1/0/6	Up	--	<not set>	<not set>	0001.64BC.DE06	
GigabitEthernet1/0/7	Up	--	<not set>	<not set>	0001.64BC.DE07	
GigabitEthernet1/0/8	Up	--	<not set>	<not set>	0001.64BC.DE08	
GigabitEthernet1/0/9	Down	1	<not set>	<not set>	0001.64BC.DE09	
GigabitEthernet1/0/10	Down	1	<not set>	<not set>	0001.64BC.DE0A	
GigabitEthernet1/0/11	Down	1	<not set>	<not set>	0001.64BC.DE0B	
GigabitEthernet1/0/12	Down	1	<not set>	<not set>	0001.64BC.DE0C	
GigabitEthernet1/0/13	Down	1	<not set>	<not set>	0001.64BC.DE0D	
GigabitEthernet1/0/14	Down	1	<not set>	<not set>	0001.64BC.DE0E	
GigabitEthernet1/0/15	Down	1	<not set>	<not set>	0001.64BC.DE0F	
GigabitEthernet1/0/16	Down	1	<not set>	<not set>	0001.64BC.DE10	
GigabitEthernet1/0/17	Down	1	<not set>	<not set>	0001.64BC.DE11	
GigabitEthernet1/0/18	Down	1	<not set>	<not set>	0001.64BC.DE12	
GigabitEthernet1/0/19	Down	1	<not set>	<not set>	0001.64BC.DE13	
GigabitEthernet1/0/20	Down	1	<not set>	<not set>	0001.64BC.DE14	
GigabitEthernet1/0/21	Down	1	<not set>	<not set>	0001.64BC.DE15	
GigabitEthernet1/0/22	Down	1	<not set>	<not set>	0001.64BC.DE16	
GigabitEthernet1/0/23	Down	1	<not set>	<not set>	0001.64BC.DE17	
GigabitEthernet1/0/24	Down	1	<not set>	<not set>	0001.64BC.DE18	
GigabitEthernet1/1/1	Down	1	<not set>	<not set>	000C.CFCD.9901	
GigabitEthernet1/1/2	Down	1	<not set>	<not set>	000C.CFCD.9902	
GigabitEthernet1/1/3	Down	1	<not set>	<not set>	000C.CFCD.9903	
GigabitEthernet1/1/4	Down	1	<not set>	<not set>	000C.CFCD.9904	
Vlan1	Down	1	<not set>	<not set>	0002.1740.C521	
Vlan10	Up	10	10.33.0.1/25	<not set>	0002.1740.C501	
Vlan20	Up	20	10.33.0.129/25	<not set>	0002.1740.C502	
Vlan30	Up	30	10.33.1.1/25	<not set>	0002.1740.C503	
Vlan40	Up	40	10.33.1.129/25	<not set>	0002.1740.C504	
Vlan50	Up	50	10.33.2.1/25	<not set>	0002.1740.C505	
Vlan60	Up	60	10.33.2.129/28	<not set>	0002.1740.C506	
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 537963_MLS1						

MLS 2

Device Name: 537963_MLS2						
Device Model: 3650-24PS						
Hostname: 537963_MLS2						
Port	Link	VLAN	IP Address	IPv6 Address	MAC Address	
GigabitEthernet1/0/1	Up	1	10.33.2.158/30	<not set>	0001.97AB.A001	
GigabitEthernet1/0/2	Up	1	10.33.2.154/30	<not set>	0001.97AB.A002	
GigabitEthernet1/0/3	Up	--	<not set>	<not set>	0001.97AB.A003	
GigabitEthernet1/0/4	Up	--	<not set>	<not set>	0001.97AB.A004	
GigabitEthernet1/0/5	Up	--	<not set>	<not set>	0001.97AB.A005	
GigabitEthernet1/0/6	Up	--	<not set>	<not set>	0001.97AB.A006	
GigabitEthernet1/0/7	Up	--	<not set>	<not set>	0001.97AB.A007	
GigabitEthernet1/0/8	Up	--	<not set>	<not set>	0001.97AB.A008	
GigabitEthernet1/0/9	Down	1	<not set>	<not set>	0001.97AB.A009	
GigabitEthernet1/0/10	Down	1	<not set>	<not set>	0001.97AB.A00A	
GigabitEthernet1/0/11	Down	1	<not set>	<not set>	0001.97AB.A00B	
GigabitEthernet1/0/12	Down	1	<not set>	<not set>	0001.97AB.A00C	
GigabitEthernet1/0/13	Down	1	<not set>	<not set>	0001.97AB.A00D	
GigabitEthernet1/0/14	Down	1	<not set>	<not set>	0001.97AB.A00E	
GigabitEthernet1/0/15	Down	1	<not set>	<not set>	0001.97AB.A00F	
GigabitEthernet1/0/16	Down	1	<not set>	<not set>	0001.97AB.A010	
GigabitEthernet1/0/17	Down	1	<not set>	<not set>	0001.97AB.A011	
GigabitEthernet1/0/18	Down	1	<not set>	<not set>	0001.97AB.A012	
GigabitEthernet1/0/19	Down	1	<not set>	<not set>	0001.97AB.A013	
GigabitEthernet1/0/20	Down	1	<not set>	<not set>	0001.97AB.A014	
GigabitEthernet1/0/21	Down	1	<not set>	<not set>	0001.97AB.A015	
GigabitEthernet1/0/22	Down	1	<not set>	<not set>	0001.97AB.A016	
GigabitEthernet1/0/23	Down	1	<not set>	<not set>	0001.97AB.A017	
GigabitEthernet1/0/24	Down	1	<not set>	<not set>	0001.97AB.A018	
GigabitEthernet1/1/1	Down	1	<not set>	<not set>	0060.3E68.E301	
GigabitEthernet1/1/2	Down	1	<not set>	<not set>	0060.3E68.E302	
GigabitEthernet1/1/3	Down	1	<not set>	<not set>	0060.3E68.E303	
GigabitEthernet1/1/4	Down	1	<not set>	<not set>	0060.3E68.E304	
Vlan1	Down	1	<not set>	<not set>	000A.4179.D71A	
Vlan10	Up	10	10.33.0.2/25	<not set>	000A.4179.D701	
Vlan20	Up	20	10.33.0.130/25	<not set>	000A.4179.D702	
Vlan30	Up	30	10.33.1.2/25	<not set>	000A.4179.D703	
Vlan40	Up	40	10.33.1.130/25	<not set>	000A.4179.D704	
Vlan50	Up	50	10.33.2.2/25	<not set>	000A.4179.D705	
Vlan60	Up	60	10.33.2.130/28	<not set>	000A.4179.D706	
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 537963_MLS2						

Switch Sales and Marketing

Device Name: 537963_Sw_SnM				
Custom Device Model: 2960 IOS15				
Hostname: 537963_Sw_SnM				
Port	Link	VLAN	IP Address	MAC Address
FastEthernet0/1	Up	--	--	0001.96D7.9A01
FastEthernet0/2	Up	--	--	0001.96D7.9A02
FastEthernet0/3	Up	10	--	0001.96D7.9A03
FastEthernet0/4	Down	10	--	0001.96D7.9A04
FastEthernet0/5	Down	10	--	0001.96D7.9A05
FastEthernet0/6	Down	10	--	0001.96D7.9A06
FastEthernet0/7	Down	10	--	0001.96D7.9A07
FastEthernet0/8	Down	10	--	0001.96D7.9A08
FastEthernet0/9	Down	10	--	0001.96D7.9A09
FastEthernet0/10	Down	10	--	0001.96D7.9A0A
FastEthernet0/11	Down	10	--	0001.96D7.9A0B
FastEthernet0/12	Down	10	--	0001.96D7.9A0C
FastEthernet0/13	Down	10	--	0001.96D7.9A0D
FastEthernet0/14	Down	10	--	0001.96D7.9A0E
FastEthernet0/15	Down	10	--	0001.96D7.9A0F
FastEthernet0/16	Down	10	--	0001.96D7.9A10
FastEthernet0/17	Down	10	--	0001.96D7.9A11
FastEthernet0/18	Down	10	--	0001.96D7.9A12
FastEthernet0/19	Down	10	--	0001.96D7.9A13
FastEthernet0/20	Down	10	--	0001.96D7.9A14
FastEthernet0/21	Down	10	--	0001.96D7.9A15
FastEthernet0/22	Down	10	--	0001.96D7.9A16
FastEthernet0/23	Down	10	--	0001.96D7.9A17
FastEthernet0/24	Down	10	--	0001.96D7.9A18
GigabitEthernet0/1	Down	1	--	0001.96D7.9A19
GigabitEthernet0/2	Down	1	--	0001.96D7.9A1A
Vlan1	Down	1	<not set>	0005.5EEB.9E01
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 537963_Sw_SnM				

Switch HR and Logistic

Device Name: 537963_Sw_HRnL				
Custom Device Model: 2960 IOS15				
Hostname: 537963_Sw_HRnL				
Port	Link	VLAN	IP Address	MAC Address
FastEthernet0/1	Up	--	--	0006.2A2D.DD01
FastEthernet0/2	Up	--	--	0006.2A2D.DD02
FastEthernet0/3	Up	20	--	0006.2A2D.DD03
FastEthernet0/4	Up	20	--	0006.2A2D.DD04
FastEthernet0/5	Down	20	--	0006.2A2D.DD05
FastEthernet0/6	Down	20	--	0006.2A2D.DD06
FastEthernet0/7	Down	20	--	0006.2A2D.DD07
FastEthernet0/8	Down	20	--	0006.2A2D.DD08
FastEthernet0/9	Down	20	--	0006.2A2D.DD09
FastEthernet0/10	Down	20	--	0006.2A2D.DD0A
FastEthernet0/11	Down	20	--	0006.2A2D.DD0B
FastEthernet0/12	Down	20	--	0006.2A2D.DD0C
FastEthernet0/13	Down	20	--	0006.2A2D.DD0D
FastEthernet0/14	Down	20	--	0006.2A2D.DD0E
FastEthernet0/15	Down	20	--	0006.2A2D.DD0F
FastEthernet0/16	Down	20	--	0006.2A2D.DD10
FastEthernet0/17	Down	20	--	0006.2A2D.DD11
FastEthernet0/18	Down	20	--	0006.2A2D.DD12
FastEthernet0/19	Down	20	--	0006.2A2D.DD13
FastEthernet0/20	Down	20	--	0006.2A2D.DD14
FastEthernet0/21	Down	20	--	0006.2A2D.DD15
FastEthernet0/22	Down	20	--	0006.2A2D.DD16
FastEthernet0/23	Down	20	--	0006.2A2D.DD17
FastEthernet0/24	Down	20	--	0006.2A2D.DD18
GigabitEthernet0/1	Down	1	--	0006.2A2D.DD19
GigabitEthernet0/2	Down	1	--	0006.2A2D.DD1A
Vlan1	Down	1	<not set>	0090.2BDD.C108
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 537963_Sw_HRnL				

Switch Finance and Administration

Device Name: 537963_Sw_FnA				
Custom Device Model: 2960 IOS15				
Hostname: 537963_Sw_FnA				
Port	Link	VLAN	IP Address	MAC Address
FastEthernet0/1	Up	--	--	0001.968B.1101
FastEthernet0/2	Up	--	--	0001.968B.1102
FastEthernet0/3	Up	30	--	0001.968B.1103
FastEthernet0/4	Down	30	--	0001.968B.1104
FastEthernet0/5	Down	30	--	0001.968B.1105
FastEthernet0/6	Down	30	--	0001.968B.1106
FastEthernet0/7	Down	30	--	0001.968B.1107
FastEthernet0/8	Down	30	--	0001.968B.1108
FastEthernet0/9	Down	30	--	0001.968B.1109
FastEthernet0/10	Down	30	--	0001.968B.110A
FastEthernet0/11	Down	30	--	0001.968B.110B
FastEthernet0/12	Down	30	--	0001.968B.110C
FastEthernet0/13	Down	30	--	0001.968B.110D
FastEthernet0/14	Down	30	--	0001.968B.110E
FastEthernet0/15	Down	30	--	0001.968B.110F
FastEthernet0/16	Down	30	--	0001.968B.1110
FastEthernet0/17	Down	30	--	0001.968B.1111
FastEthernet0/18	Down	30	--	0001.968B.1112
FastEthernet0/19	Down	30	--	0001.968B.1113
FastEthernet0/20	Down	30	--	0001.968B.1114
FastEthernet0/21	Down	30	--	0001.968B.1115
FastEthernet0/22	Down	30	--	0001.968B.1116
FastEthernet0/23	Down	30	--	0001.968B.1117
FastEthernet0/24	Down	30	--	0001.968B.1118
GigabitEthernet0/1	Down	1	--	0001.968B.1119
GigabitEthernet0/2	Down	1	--	0001.968B.111A
Vlan1	Down	1	<not set>	0007.EC71.B479
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 537963_Sw_FnA				

Switch Admin and Humas

Device Name: 537963_Sw_AnH				
Custom Device Model: 2960 IOS15				
Hostname: 537963_Sw_AnH				
Port	Link	VLAN	IP Address	MAC Address
FastEthernet0/1	Up	--	--	000C.8529.E201
FastEthernet0/2	Up	--	--	000C.8529.E202
FastEthernet0/3	Up	40	--	000C.8529.E203
FastEthernet0/4	Down	40	--	000C.8529.E204
FastEthernet0/5	Down	40	--	000C.8529.E205
FastEthernet0/6	Down	40	--	000C.8529.E206
FastEthernet0/7	Down	40	--	000C.8529.E207
FastEthernet0/8	Down	40	--	000C.8529.E208
FastEthernet0/9	Down	40	--	000C.8529.E209
FastEthernet0/10	Down	40	--	000C.8529.E20A
FastEthernet0/11	Down	40	--	000C.8529.E20B
FastEthernet0/12	Down	40	--	000C.8529.E20C
FastEthernet0/13	Down	40	--	000C.8529.E20D
FastEthernet0/14	Down	40	--	000C.8529.E20E
FastEthernet0/15	Down	40	--	000C.8529.E20F
FastEthernet0/16	Down	40	--	000C.8529.E210
FastEthernet0/17	Down	40	--	000C.8529.E211
FastEthernet0/18	Down	40	--	000C.8529.E212
FastEthernet0/19	Down	40	--	000C.8529.E213
FastEthernet0/20	Down	40	--	000C.8529.E214
FastEthernet0/21	Down	40	--	000C.8529.E215
FastEthernet0/22	Down	40	--	000C.8529.E216
FastEthernet0/23	Down	40	--	000C.8529.E217
FastEthernet0/24	Down	40	--	000C.8529.E218
GigabitEthernet0/1	Down	1	--	000C.8529.E219
GigabitEthernet0/2	Down	1	--	000C.8529.E21A
Vlan1	Down	1	<not set>	00E0.F7B4.66B7
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 537963_Sw_AnH				

Switch IT

Device Name: 537963_Sw_IT				
Custom Device Model: 2960 IOS15				
Hostname: 537963_Sw_IT				
Port	Link	VLAN	IP Address	MAC Address
FastEthernet0/1	Up	--	--	0001.63CD.0C01
FastEthernet0/2	Up	--	--	0001.63CD.0C02
FastEthernet0/3	Up	50	--	0001.63CD.0C03
FastEthernet0/4	Down	50	--	0001.63CD.0C04
FastEthernet0/5	Down	50	--	0001.63CD.0C05
FastEthernet0/6	Down	50	--	0001.63CD.0C06
FastEthernet0/7	Down	50	--	0001.63CD.0C07
FastEthernet0/8	Down	50	--	0001.63CD.0C08
FastEthernet0/9	Down	50	--	0001.63CD.0C09
FastEthernet0/10	Down	50	--	0001.63CD.0C0A
FastEthernet0/11	Down	50	--	0001.63CD.0C0B
FastEthernet0/12	Down	50	--	0001.63CD.0C0C
FastEthernet0/13	Down	50	--	0001.63CD.0C0D
FastEthernet0/14	Down	50	--	0001.63CD.0C0E
FastEthernet0/15	Down	50	--	0001.63CD.0C0F
FastEthernet0/16	Down	50	--	0001.63CD.0C10
FastEthernet0/17	Down	50	--	0001.63CD.0C11
FastEthernet0/18	Down	50	--	0001.63CD.0C12
FastEthernet0/19	Down	50	--	0001.63CD.0C13
FastEthernet0/20	Down	50	--	0001.63CD.0C14
FastEthernet0/21	Down	50	--	0001.63CD.0C15
FastEthernet0/22	Down	50	--	0001.63CD.0C16
FastEthernet0/23	Down	50	--	0001.63CD.0C17
FastEthernet0/24	Down	50	--	0001.63CD.0C18
GigabitEthernet0/1	Down	1	--	0001.63CD.0C19
GigabitEthernet0/2	Down	1	--	0001.63CD.0C1A
Vlan1	Down	1	<not set>	0060.4754.5EAC
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 537963_Sw_IT				

Switch Server

Device Name: 537963_Sw_Server				
Custom Device Model: 2960 IOS15				
Hostname: 537963_Sw_Server				
Port	Link	VLAN	IP Address	MAC Address
FastEthernet0/1	Up	--	--	0090.2BD3.7C01
FastEthernet0/2	Up	--	--	0090.2BD3.7C02
FastEthernet0/3	Up	60	--	0090.2BD3.7C03
FastEthernet0/4	Up	60	--	0090.2BD3.7C04
FastEthernet0/5	Down	60	--	0090.2BD3.7C05
FastEthernet0/6	Down	60	--	0090.2BD3.7C06
FastEthernet0/7	Down	60	--	0090.2BD3.7C07
FastEthernet0/8	Down	60	--	0090.2BD3.7C08
FastEthernet0/9	Down	60	--	0090.2BD3.7C09
FastEthernet0/10	Down	60	--	0090.2BD3.7C0A
FastEthernet0/11	Down	60	--	0090.2BD3.7C0B
FastEthernet0/12	Down	60	--	0090.2BD3.7C0C
FastEthernet0/13	Down	60	--	0090.2BD3.7C0D
FastEthernet0/14	Down	60	--	0090.2BD3.7C0E
FastEthernet0/15	Down	60	--	0090.2BD3.7C0F
FastEthernet0/16	Down	60	--	0090.2BD3.7C10
FastEthernet0/17	Down	60	--	0090.2BD3.7C11
FastEthernet0/18	Down	60	--	0090.2BD3.7C12
FastEthernet0/19	Down	60	--	0090.2BD3.7C13
FastEthernet0/20	Down	60	--	0090.2BD3.7C14
FastEthernet0/21	Down	60	--	0090.2BD3.7C15
FastEthernet0/22	Down	60	--	0090.2BD3.7C16
FastEthernet0/23	Down	60	--	0090.2BD3.7C17
FastEthernet0/24	Down	60	--	0090.2BD3.7C18
GigabitEthernet0/1	Down	1	--	0090.2BD3.7C19
GigabitEthernet0/2	Down	1	--	0090.2BD3.7C1A
Vlan1	Down	1	<not set>	0060.7020.BBBC
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 537963_Sw_Server				

PC Sales and Marketing

Device Name: 537963_PC_SnM				
Device Model: PC-PT				
Port	Link	IP Address	IPv6 Address	MAC Address
FastEthernet0	Up	10.33.0.3/25	<not set>	0001.43E7.05AE
Bluetooth	Down	<not set>	<not set>	00E0.F7A1.354E
Gateway: 10.33.0.1				
DNS Server: 10.33.2.131				
Line Number: <not set>				
Physical Location: Intercity > Home City > Corporate Office > 537963_PC_SnM				

PC HR and Logistic

Device Name: 537963_PC_HRnL				
Device Model: PC-PT				
Port	Link	IP Address	IPv6 Address	MAC Address
FastEthernet0	Up	10.33.0.131/25	<not set>	0007.EC20.DA66
Bluetooth	Down	<not set>	<not set>	0004.9AC7.1310
Gateway: 10.33.0.129				
DNS Server: 10.33.2.121				
Line Number: <not set>				
Physical Location: Intercity > Home City > Corporate Office > 537963_PC_HRnL				

PC Finance and Administration

Device Name: 537963_PC_FnA				
Device Model: PC-PT				
Port	Link	IP Address	IPv6 Address	MAC Address
FastEthernet0	Up	10.33.1.3/25	<not set>	000C.CF4C.602B
Bluetooth	Down	<not set>	<not set>	0002.170E.3670
Gateway: 10.33.1.1				
DNS Server: 10.33.2.131				
Line Number: <not set>				
Physical Location: Intercity > Home City > Corporate Office > 537963_PC_FnA				

PC Admin and Humas

Device Name: 537963_PC_AnH				
Device Model: PC-PT				
Port	Link	IP Address	IPv6 Address	MAC Address
FastEthernet0	Up	10.33.1.131/25	<not set>	0030.F2A0.A77A
Bluetooth	Down	<not set>	<not set>	0002.4A61.2B7E
Gateway: 10.33.1.129				
DNS Server: 10.33.2.131				
Line Number: <not set>				
Physical Location: Intercity > Home City > Corporate Office > 537963_PC_AnH				

PC IT

Device Name: 537963_PC_IT				
Device Model: PC-PT				
Port	Link	IP Address	IPv6 Address	MAC Address
FastEthernet0	Up	10.33.2.3/25	<not set>	0090.2BB4.C4EA
Bluetooth	Down	<not set>	<not set>	0060.4703.8580
Gateway: 10.33.2.1				
DNS Server: 10.33.2.131				
Line Number: <not set>				
Physical Location: Intercity > Home City > Corporate Office > 537963_PC_IT				

Server DNS

```

Device Name: 537963_DNS Server
Device Model: Server-PT

Port      Link    IP Address    IPv6 Address    MAC Address
FastEthernet0  Up    10.33.2.131/28    <not set>    00D0.BC55.13E6

Gateway: 10.33.2.129
DNS Server: 10.33.2.131
Line Number: <not set>

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 537963_DNS Server

```

Server DHCP

```

Device Name: 537963_DHCP Server
Device Model: Server-PT

Port      Link    IP Address    IPv6 Address    MAC Address
FastEthernet0  Up    10.33.2.133/28    <not set>    00D0.58D4.5624

Gateway: 10.33.2.129
DNS Server: 10.33.2.131
Line Number: <not set>

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 537963_DHCP Server

```

Access Point

```

Device Name: 537963_AP_HRnL
Device Model: AccessPoint-PT

Port      Link
Port 0    Up
Port 1    Up

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 537963_AP_HRnL

```

Laptop

```

Device Name: 537963_Laptop
Device Model: Laptop-PT

Port      Link    IP Address    IPv6 Address    MAC Address
Wireless0  Up    10.33.0.134/25    <not set>    0030.A398.B0B1
Bluetooth  Down    <not set>    <not set>    0060.2FE8.019E

Gateway: 10.33.0.129
DNS Server: 10.33.2.121
Line Number: <not set>

Wireless Best Data Rate: 54 Mbps
Wireless Signal Strength: 67%

Physical Location: Intercity > Home City > Corporate Office > 537963_Laptop

```

6. Verifikasi routing

ISP 1


```

537963_ISP1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
O       10.33.0.0/25 [110/66] via 205.36.19.2, 00:14:25, Serial0/3/0
O       10.33.0.128/25 [110/66] via 205.36.19.2, 00:14:25, Serial0/3/0
O       10.33.1.0/25 [110/66] via 205.36.19.2, 00:14:25, Serial0/3/0
O       10.33.1.128/25 [110/66] via 205.36.19.2, 00:14:25, Serial0/3/0
O       10.33.2.0/25 [110/66] via 205.36.19.2, 00:14:25, Serial0/3/0
O       10.33.2.128/28 [110/66] via 205.36.19.2, 00:14:25, Serial0/3/0
O       10.33.2.144/30 [110/65] via 205.36.19.2, 00:14:40, Serial0/3/0
O       10.33.2.148/30 [110/66] via 205.36.19.2, 00:14:25, Serial0/3/0
O       10.33.2.152/30 [110/65] via 205.36.19.2, 00:14:25, Serial0/3/0
O       10.33.2.156/30 [110/66] via 205.36.19.2, 00:14:25, Serial0/3/0
    205.36.19.0/24 is variably subnetted, 6 subnets, 2 masks
C       205.36.19.0/30 is directly connected, Serial0/3/0
L       205.36.19.1/32 is directly connected, Serial0/3/0
O       205.36.19.4/30 [110/128] via 205.36.19.2, 00:15:15, Serial0/3/0
C       205.36.19.8/30 is directly connected, Serial0/3/1
L       205.36.19.9/32 is directly connected, Serial0/3/1
O       205.36.19.12/30 [110/130] via 205.36.19.2, 00:14:25, Serial0/3/0

```

ISP 2

```

537963_ISP2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    205.36.19.0/24 is variably subnetted, 4 subnets, 2 masks
C       205.36.19.4/30 is directly connected, Serial0/3/0
L       205.36.19.5/32 is directly connected, Serial0/3/0
C       205.36.19.12/30 is directly connected, Serial0/3/1
L       205.36.19.13/32 is directly connected, Serial0/3/1

```

Router 1


```

537963_RT1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 12 subnets, 4 masks
O       10.33.0.0/25 [110/2] via 10.33.2.146, 00:15:16, GigabitEthernet0/0
        [110/2] via 10.33.2.154, 00:15:16, GigabitEthernet0/1
O       10.33.0.128/25 [110/2] via 10.33.2.146, 00:15:16, GigabitEthernet0/0
        [110/2] via 10.33.2.154, 00:15:16, GigabitEthernet0/1
O       10.33.1.0/25 [110/2] via 10.33.2.146, 00:15:16, GigabitEthernet0/0
        [110/2] via 10.33.2.154, 00:15:16, GigabitEthernet0/1
O       10.33.1.128/25 [110/2] via 10.33.2.146, 00:15:16, GigabitEthernet0/0
        [110/2] via 10.33.2.154, 00:15:16, GigabitEthernet0/1
O       10.33.2.0/25 [110/2] via 10.33.2.146, 00:15:16, GigabitEthernet0/0
        [110/2] via 10.33.2.154, 00:15:16, GigabitEthernet0/1
O       10.33.2.128/28 [110/2] via 10.33.2.146, 00:15:16, GigabitEthernet0/0
        [110/2] via 10.33.2.154, 00:15:16, GigabitEthernet0/1
C       10.33.2.144/30 is directly connected, GigabitEthernet0/0
L       10.33.2.145/32 is directly connected, GigabitEthernet0/0
O       10.33.2.148/30 [110/2] via 10.33.2.146, 00:15:31, GigabitEthernet0/0
C       10.33.2.152/30 is directly connected, GigabitEthernet0/1
L       10.33.2.153/32 is directly connected, GigabitEthernet0/1
O       10.33.2.156/30 [110/2] via 10.33.2.154, 00:15:16, GigabitEthernet0/1
    205.36.19.0/24 is variably subnetted, 6 subnets, 2 masks
C       205.36.19.0/30 is directly connected, Serial0/3/0
L       205.36.19.2/32 is directly connected, Serial0/3/0
C       205.36.19.4/30 is directly connected, Serial0/3/1
L       205.36.19.6/32 is directly connected, Serial0/3/1
O       205.36.19.8/30 [110/66] via 10.33.2.146, 00:15:16, GigabitEthernet0/0
        [110/66] via 10.33.2.154, 00:15:16, GigabitEthernet0/1
O       205.36.19.12/30 [110/66] via 10.33.2.146, 00:15:16, GigabitEthernet0/0
        [110/66] via 10.33.2.154, 00:15:16, GigabitEthernet0/1

```

Router 2

```

537963_RT2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 12 subnets, 4 masks
O       10.33.0.0/25 [110/2] via 10.33.2.158, 00:15:56, GigabitEthernet0/0
        [110/2] via 10.33.2.149, 00:15:56, GigabitEthernet0/1
O       10.33.0.128/25 [110/2] via 10.33.2.158, 00:15:56, GigabitEthernet0/0
        [110/2] via 10.33.2.149, 00:15:56, GigabitEthernet0/1
O       10.33.1.0/25 [110/2] via 10.33.2.158, 00:15:56, GigabitEthernet0/0
        [110/2] via 10.33.2.149, 00:15:56, GigabitEthernet0/1
O       10.33.1.128/25 [110/2] via 10.33.2.158, 00:15:56, GigabitEthernet0/0
        [110/2] via 10.33.2.149, 00:15:56, GigabitEthernet0/1
O       10.33.2.0/25 [110/2] via 10.33.2.158, 00:15:56, GigabitEthernet0/0
        [110/2] via 10.33.2.149, 00:15:56, GigabitEthernet0/1
O       10.33.2.128/28 [110/2] via 10.33.2.158, 00:15:56, GigabitEthernet0/0
        [110/2] via 10.33.2.149, 00:15:56, GigabitEthernet0/1
O       10.33.2.144/30 [110/2] via 10.33.2.149, 00:15:56, GigabitEthernet0/1
C       10.33.2.148/30 is directly connected, GigabitEthernet0/1
L       10.33.2.150/32 is directly connected, GigabitEthernet0/1
O       10.33.2.152/30 [110/2] via 10.33.2.158, 00:15:56, GigabitEthernet0/0
C       10.33.2.156/30 is directly connected, GigabitEthernet0/0
L       10.33.2.157/32 is directly connected, GigabitEthernet0/0
    205.36.19.0/24 is variably subnetted, 6 subnets, 2 masks
O       205.36.19.0/30 [110/66] via 10.33.2.158, 00:15:56, GigabitEthernet0/0
        [110/66] via 10.33.2.149, 00:15:56, GigabitEthernet0/1
O       205.36.19.4/30 [110/66] via 10.33.2.158, 00:15:56, GigabitEthernet0/0
        [110/66] via 10.33.2.149, 00:15:56, GigabitEthernet0/1
C       205.36.19.8/30 is directly connected, Serial0/3/0
L       205.36.19.10/32 is directly connected, Serial0/3/0
C       205.36.19.12/30 is directly connected, Serial0/3/1
L       205.36.19.14/32 is directly connected, Serial0/3/1

```

MLS 1

```
537963_MLS1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
C       10.33.0.0/25 is directly connected, Vlan10
C       10.33.0.128/25 is directly connected, Vlan20
C       10.33.1.0/25 is directly connected, Vlan30
C       10.33.1.128/25 is directly connected, Vlan40
C       10.33.2.0/25 is directly connected, Vlan50
C       10.33.2.128/28 is directly connected, Vlan60
C       10.33.2.144/30 is directly connected, GigabitEthernet1/0/1
C       10.33.2.148/30 is directly connected, GigabitEthernet1/0/2
O       10.33.2.152/30 [110/2] via 10.33.2.145, 00:16:35, GigabitEthernet1/0/1
                        [110/2] via 10.33.0.2, 00:16:35, Vlan10
                        [110/2] via 10.33.0.130, 00:16:35, Vlan20
                        [110/2] via 10.33.1.2, 00:16:35, Vlan30
                        [110/2] via 10.33.1.130, 00:16:35, Vlan40
                        [110/2] via 10.33.2.2, 00:16:35, Vlan50
                        [110/2] via 10.33.2.130, 00:16:35, Vlan60
O       10.33.2.156/30 [110/2] via 10.33.2.150, 00:16:35, GigabitEthernet1/0/2
                        [110/2] via 10.33.0.2, 00:16:35, Vlan10
                        [110/2] via 10.33.0.130, 00:16:35, Vlan20
                        [110/2] via 10.33.1.2, 00:16:35, Vlan30
                        [110/2] via 10.33.1.130, 00:16:35, Vlan40
                        [110/2] via 10.33.2.2, 00:16:35, Vlan50
                        [110/2] via 10.33.2.130, 00:16:35, Vlan60
    205.36.19.0/30 is subnetted, 4 subnets
O       205.36.19.0 [110/65] via 10.33.2.145, 00:16:45, GigabitEthernet1/0/1
O       205.36.19.4 [110/65] via 10.33.2.145, 00:16:45, GigabitEthernet1/0/1
O       205.36.19.8 [110/65] via 10.33.2.150, 00:16:45, GigabitEthernet1/0/2
O       205.36.19.12 [110/65] via 10.33.2.150, 00:16:45, GigabitEthernet1/0/2
S*    0.0.0.0/0 is directly connected, GigabitEthernet1/0/1
        is directly connected, GigabitEthernet1/0/2
```

MLS 2

```

537963_MLS2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
C       10.33.0.0/25 is directly connected, Vlan10
C       10.33.0.128/25 is directly connected, Vlan20
C       10.33.1.0/25 is directly connected, Vlan30
C       10.33.1.128/25 is directly connected, Vlan40
C       10.33.2.0/25 is directly connected, Vlan50
C       10.33.2.128/28 is directly connected, Vlan60
O       10.33.2.144/30 [110/2] via 10.33.2.153, 00:17:42, GigabitEthernet1/0/2
                        [110/2] via 10.33.0.1, 00:17:42, Vlan10
                        [110/2] via 10.33.0.129, 00:17:42, Vlan20
                        [110/2] via 10.33.1.1, 00:17:42, Vlan30
                        [110/2] via 10.33.1.129, 00:17:42, Vlan40
                        [110/2] via 10.33.2.1, 00:17:42, Vlan50
                        [110/2] via 10.33.2.129, 00:17:42, Vlan60
O       10.33.2.148/30 [110/2] via 10.33.2.157, 00:17:42, GigabitEthernet1/0/1
                        [110/2] via 10.33.0.1, 00:17:42, Vlan10
                        [110/2] via 10.33.0.129, 00:17:42, Vlan20
                        [110/2] via 10.33.1.1, 00:17:42, Vlan30
                        [110/2] via 10.33.1.129, 00:17:42, Vlan40
                        [110/2] via 10.33.2.1, 00:17:42, Vlan50
                        [110/2] via 10.33.2.129, 00:17:42, Vlan60
C       10.33.2.152/30 is directly connected, GigabitEthernet1/0/2
C       10.33.2.156/30 is directly connected, GigabitEthernet1/0/1
205.36.19.0/30 is subnetted, 4 subnets
O       205.36.19.0 [110/65] via 10.33.2.153, 00:17:42, GigabitEthernet1/0/2
O       205.36.19.4 [110/65] via 10.33.2.153, 00:17:42, GigabitEthernet1/0/2
O       205.36.19.8 [110/65] via 10.33.2.157, 00:17:42, GigabitEthernet1/0/1
O       205.36.19.12 [110/65] via 10.33.2.157, 00:17:42, GigabitEthernet1/0/1
S*    0.0.0.0/0 is directly connected, GigabitEthernet1/0/1
       is directly connected, GigabitEthernet1/0/2

```

7. Verifikasi VLAN (screenshot pada masing-masing multilayer switch)

MLS 1

```

Ini MLS1, no unauthorized access!

537963_MLS1>en
537963_MLS1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
537963_MLS1(config)#do show vlan br

VLAN Name                Status    Ports
-----
1    default                active    Gig1/0/9, Gig1/0/10, Gig1/0/11, Gig1/0/12
                                Gig1/0/13, Gig1/0/14, Gig1/0/15, Gig1/0/16
                                Gig1/0/17, Gig1/0/18, Gig1/0/19, Gig1/0/20
                                Gig1/0/21, Gig1/0/22, Gig1/0/23, Gig1/0/24
                                Gig1/1/1, Gig1/1/2, Gig1/1/3, Gig1/1/4
10   SnM                    active
20   HRnL                   active
30   FnA                    active
40   AnH                    active
50   IT                     active
60   Server                 active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
537963_MLS1(config)#

```

MLS 2

```

Ini MLS2, no unauthorized access!

537963_MLS2>en
537963_MLS2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
537963_MLS2(config)#do show vlan br

```

VLAN	Name	Status	Ports
1	default	active	Gig1/0/9, Gig1/0/10, Gig1/0/11, Gig1/0/12, Gig1/0/13, Gig1/0/14, Gig1/0/15, Gig1/0/16, Gig1/0/17, Gig1/0/18, Gig1/0/19, Gig1/0/20, Gig1/0/21, Gig1/0/22, Gig1/0/23, Gig1/0/24, Gig1/1/1, Gig1/1/2, Gig1/1/3, Gig1/1/4
10	SnM	active	
20	HRnL	active	
30	FnA	active	
40	AnH	active	
50	IT	active	
60	Server	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```

537963_MLS2(config)#

```

8. Verifikasi DHCP

PC Sales and Marketing

537963_PC_SnM

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address: 10.33.0.3

Subnet Mask: 255.255.255.128

Default Gateway: 10.33.0.1

DNS Server: 10.33.2.131

PC HR and Logistic

537963_PC_HRnL

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static

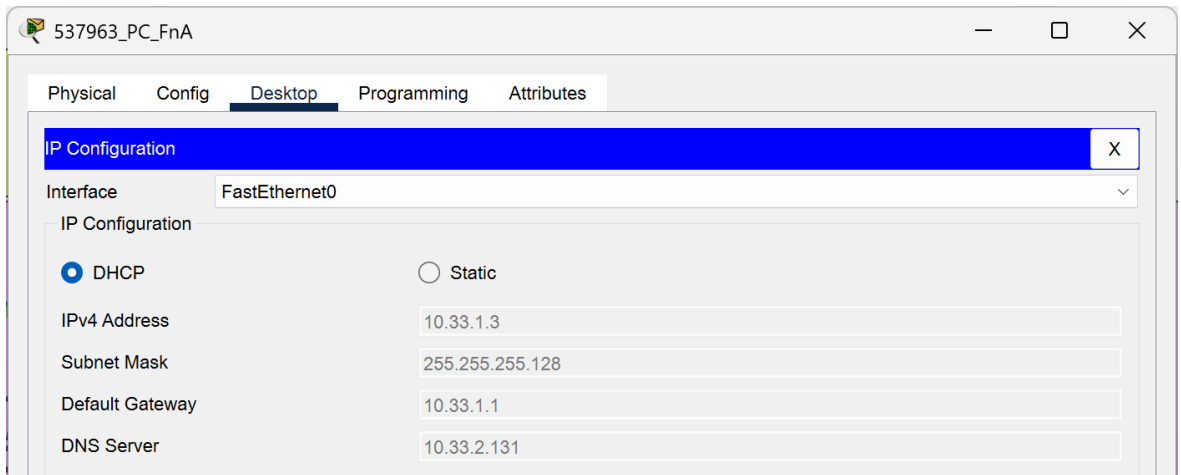
IPv4 Address: 10.33.0.131

Subnet Mask: 255.255.255.128

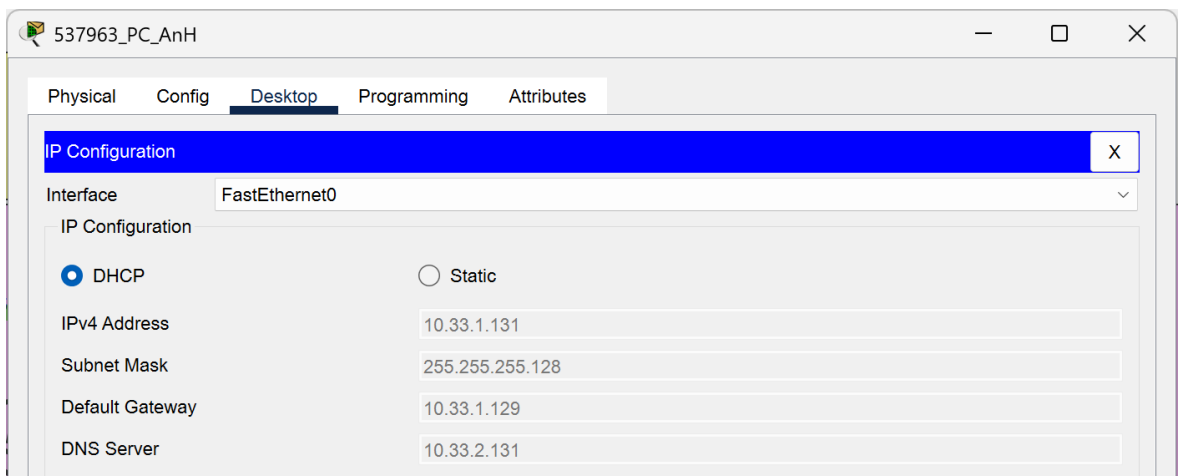
Default Gateway: 10.33.0.129

DNS Server: 10.33.2.121

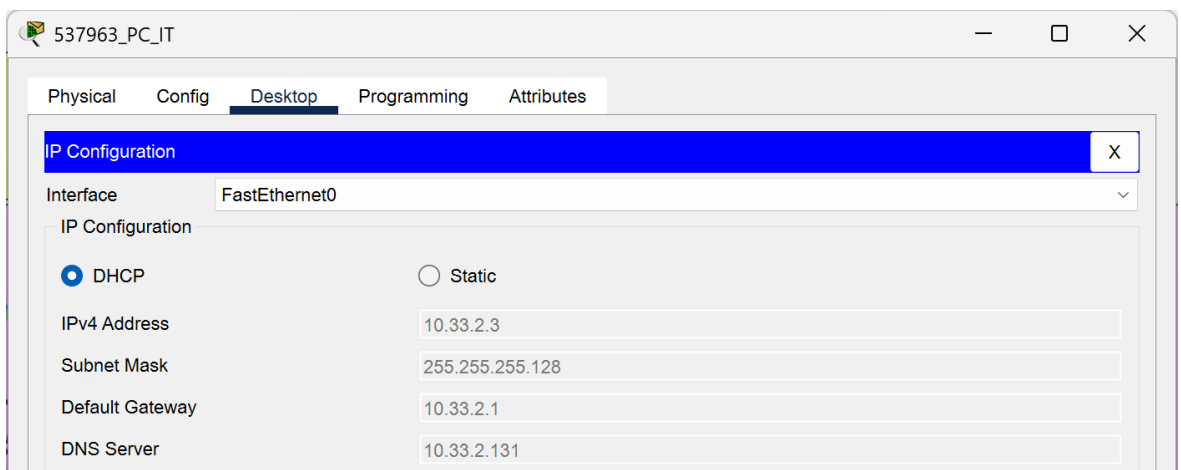
PC Finance and Administration



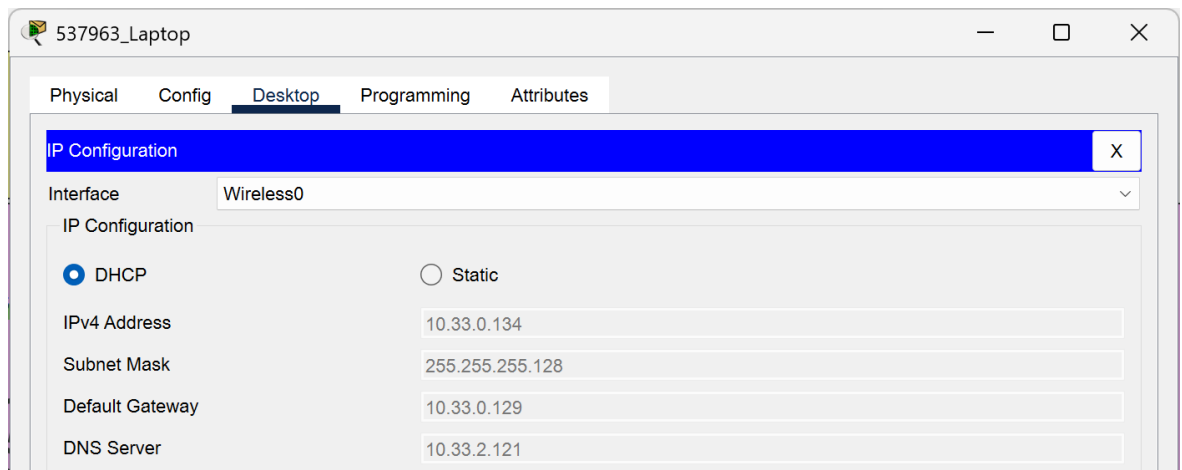
PC Admin and Humas



PC IT



Laptop



PEMBAHASAN

Pada praktikum kali ini, dilakukan perancangan dan konfigurasi infrastruktur jaringan untuk gedung baru Perusahaan Nagoya. Jaringan ini dibangun menggunakan arsitektur hierarkis yang melibatkan dua multilayer switch, dua router internal, dan terhubung ke dua ISP untuk memastikan ketersediaan jalur cadangan (redundancy). Berdasarkan pengujian, berikut tahapannya:

- Implementasi VLAN dan Inter-VLAN Routing
Jaringan dibagi secara logis menggunakan VLAN 10 hingga 60 untuk memisahkan lalu lintas dari enam departemen yang berbeda. Agar antar-departemen dapat saling berkomunikasi, diaktifkan fitur Inter-VLAN Routing pada multilayer switch.
- DHCP Server dan IP Helper
- Karena layanan DHCP Server berpusat di ruang server, dilakukan konfigurasi ip helper-address pada multilayer switch sebagai perantara. Hasil pengujian menunjukkan PC klien di tiap VLAN berhasil mendapatkan IP secara otomatis sesuai dengan porsi subnet masing-masing.
- Routing OSPF
Untuk menghubungkan seluruh perangkat, diterapkan protokol routing dinamis OSPF pada multilayer switch, router internal, dan router ISP. Menggunakan perintah show ip route, rute OSPF terpantau berjalan dengan baik (ditandai dengan kode 'O').
- Keamanan Jaringan (SSH dan Port-Security)

Untuk keamanan manajemen jarak jauh, dilakukan konfigurasi SSH versi 2 pada setiap router dan multilayer switch. Pada sisi keamanan akses fisik, diaktifkan switchport security pada departemen Finansial dan Akunting dengan metode sticky dan violation shutdown untuk membatasi akses pada perangkat yang tidak sah.

- Uji Konektivitas dan Jaringan Wireless
Kebutuhan konektivitas wireless dipenuhi dengan pemasangan Access Point menggunakan autentikasi keamanan WPA2-PSK di departemen HR & Logistik. Berdasarkan pengujian ping silang antar-perangkat dan departemen, koneksi kabel maupun wireless terbukti berjalan sukses tanpa ada loss.

DAFTAR PUSTAKA

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